TIPA Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

Proposer	Name	Dr. Felix HEUBNER	Email	Felix.Heubner@ifam-dd.fraunhofer.de
Institute Name	Fraunhofer Institute for Manufacturing Technology and Advanced Materials, IFAM			
Address	Winterbergstrasse 28, 01277 Dresden, Germany			

2. Technology Overview

Title of Proposed Technology	*Development of Ni Foam–Based Porous Transport Layer (PTL) with Advanced Surface Coatings for AEL/AEM Electrolysis*			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen (√), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Energy & Resources		*Mechanical・Materials*	
	Medium Category	Small Category	Medium Category	Small Category
	New & Renewable Energy	*Hydrogen*	*Surface Treatment*	*Coating/Method Technologies*
Concept of Proposed Technology	The proposed technology targets a high-performance nickel foam-based porous transport layer (PTL) / electrode substrate for alkaline electrolysis (AEL) and alkaline membrane electrolysis (AEM). It combines optimized 3D Ni foam architectures with scalable, uniform surface coatings (PGM-free, Ni-based) to reduce interfacial contact resistance, improve gas and water transport, and enhance electrochemical durability. The developed PTLs will be validated in AEL/AEM cell and short-stack environments to support low-cost, large-scale green hydrogen production.			
Rationale for Joint R&D Project	Although nickel foam offers high conductivity and alkaline stability, conventional PTLs/electrode substrates often show non-uniform surface activity, elevated interfacial contact resistance, and degradation at high current density. Solving these issues requires combining industrial Ni foam manufacturing and scale-up with advanced coating, electrochemical testing, and durability expertise. This joint R&D will link a Korean SME's substrate and pilot-production capabilities with Fraunhofer			

(max 10 pages)

	IFAM's strengths in surface engineering and electrochemical validation to accelerate translation from lab concepts to manufacturable PTL prototypes for AEL/AEM stacks.
Key Words	1. porous transport layer (PTL) 2. AEL/AEM electrolysis 3. Interfacial contact resistance (ICR) 4. surface coating technology 5.Green Hydrogen production

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	**Start of the Project**	Level (3)	**End of the Project**	Level (5)
R&D Objectives	The objective of this R&D project is to develop and validate a high-performance Ni foam-based porous transport layer (PTL) or electrode substrate for AEL and AEM electrolysis with low contact resistance, high electrochemical durability, and industrial scalability. The work includes optimization of Ni foam structural parameters and compression behavior, development of scalable PGM-free Ni-based surface coatings and microstructured layers, and electrochemical validation at cell and short-stack level. The project will additionally demonstrate pilot-scale manufacturability and deliver prototype PTLs suitable for TRL 6 validation in representative AEL/AEM stack environments.			
R&D Target Specifications	The project targets a reduction of interfacial contact resistance by 30% compared to state-of-the-art uncoated Ni foam PTLs, combined with stable electrochemical performance at industrially relevant current densities (1 A/cm²). The developed PTLs shall demonstrate durability under prolonged operation (<=50 mV / 1000h) and accelerated stress testing in alkaline environments, with no critical degradation of electrical or structural properties. In addition, the coating processes and Ni foam designs shall be compatible with pilot-scale manufacturing, enabling reproducible quality, uniform coating coverage on three-dimensional structures, and readiness for prototype-level validation at TRL 6.			
R&D Plan	The R&D activities are organized into five work packages (WP1-WP5) spanning Phase 1 (6 months) and Phase 2 (30 months). Phase 1 focuses on substrate down-selection, coating feasibility, and electrochemical proof-of-concept, while Phase 2 addresses optimization, durability validation, and pilot-scale manufacturability, concluding with TRL 6 pilot-scale prototype validation for AEL and AEM electrolyzer stacks. Phase 1 - Pre-study (Months 1-6): WP1 - Ni Foam Substrate Design and Characterization Timeline: Months 1-3 WP1 establishes the material and structural foundation of the PTL development. Nickel foam substrates will be systematically selected and screened within practical commercial ranges (pore size 300-1200 um, porosity 80-95%, thickness 0.8-2.0 mm). Mechanical stability, electrical conductivity,			

	and contact behavior under compression will be characterized under electrolysis-relevant conditions. Structure-property relationships will be derived to define suitable substrate windows for coating and electrochemical testing. Milestone: Selection of optimized Ni foam substrate designs for coating development. WP2 - Surface Treatment and Coating Feasibility, Timeline: Months 2-4 In WP2, PGM-free Ni-based surface coatings will be developed and applied to selected Ni foam substrates. The focus is on achieving uniform 3D coverage, good adhesion, and corrosion resistance using scalable coating approaches. Initial screening of homogeneous and graded coating concepts will be performed to identify promising candidates for electrochemical validation. Milestone: Feasible coating concepts demonstrated on Ni foam substrates. WP3 - Electrochemical Screening and Validation, Timeline: Months 4-6 WP3 evaluates the electrochemical performance of coated and uncoated PTLs in AEL and AEM single-cell environments. Key metrics include interfacial contact resistance (ICR), EIS, and high-frequency resistance (HFR). Gas evolution and mass transport behavior will be assessed to confirm performance improvements relative to baseline PTLs. Initial durability screening and definition of pilot-process requirements will be performed. Phase 1 Milestone: Proof-of-concept PTL materials demonstrating improved electrochemical performance (TRL 4). Phase 2 - Main R&D (Months 7-36): WP1 (continued) - Advanced Substrate Optimization, Timeline: Months 7-12 Based on Phase 1 results, Ni foam architectures will be further optimized with respect to pore hierarchy, mechanical robustness, and electrical pathways. Design refinements will target improved interaction with coating layers and enhanced mass transport under high current density operation. Milestone: Finalized Ni foam substrate design for long-term validation. WP2 (continued) - Advanced Coating Development and Scale Compatibility, Timeline: Months 10-20 Coating compositions and deposition processes will be optimized for durability, low contact resistance, and manufacturability. Penetration depth, coating thickness, and gradient concepts will be refined to ensure reproducibility and compatibility with pilot-scale processing. Milestone: Optimized coating process compatible with pilot-scale production. WP4 - Durability and Degradation Analysis, Timeline: Months 18-30 WP4 focuses on long-term stability using accelerated stress testing (AST) and extended alkaline corrosion tests under electrolysis-relevant conditions. Degradation mechanisms will be analyzed using SEM/EDX, XRD, and ICP, providing feedback for final material adjustments. Milestone: Demonstrated durability and identified lifetime-limiting mechanisms. WP5 - Pilot Manufacturing and Prototype Validation, Timeline: Months 28-36

	In WP5, optimized PTL designs will be transferred to pilot-scale manufacturing. Quality specifications, inspection protocols, and reproducibility criteria will be established. Prototype PTLs will be validated in representative AEL and AEM stack environments for performance evaluation of finalized components. Final Milestone: TRL 6 achieved - pilot-scale production and performance validation of Ni foam-based PTL prototypes enabling low-cost green hydrogen production.

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	Fraunhofer IFAM will contribute its expertise in advanced metallic materials, surface engineering, and electrochemical technology to develop and optimize Ni foam–based PTL concepts for AEL and AEM electrolysis. The institute will be responsible for surface coating process development, electrochemical testing and validation at cell and short-stack level, as well as durability, corrosion, and degradation mechanism analysis under realistic operating conditions. In addition, Fraunhofer IFAM will provide technical feedback for structural and process optimization, supporting the transition from laboratory-scale concepts to pilot-scale manufacturable PTL solutions.
Expected R&R of Korean SME Partner	The Korean SME partner will be responsible for the design, sourcing, and manufacturing of Ni foam PTL substrates, including control of pore structure, porosity, thickness, and mechanical properties. The SME will lead process industrialization, quality control, and pilot-scale production, ensuring that the developed PTL designs are compatible with industrial manufacturing. Furthermore, the SME will prepare prototype and customer-ready PTL samples and drive commercialization and market deployment, enabling the integration of the developed technology into AEL and AEM electrolyzer stacks for green hydrogen production.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	This joint R&D project is expected to deliver a validated high-performance Ni foam–based porous transport layer (PTL) for application in alkaline and alkaline membrane electrolysis systems. The developed PTLs will demonstrate reduced interfacial contact resistance, improved electrolysis efficiency, and enhanced durability, enabling longer operational lifetimes under industrially relevant conditions. From an industrial perspective, the project will support the commercial readiness and scale-up of advanced hydrogen components, strengthening the global competitiveness of the Korean hydrogen supply chain. In addition, the collaboration will establish a sustainable international R&D partnership, bridging applied research and industrial manufacturing, accelerating commercialization, and contributing to the cost-effective production of green hydrogen and Korea's position in the global hydrogen value chain.

(max 10 pages)

6. References

(1) (1) Yang, Y.; Li, P.; Zheng, X.; Sun, W.; Dou, S. X.; Ma, T.; Pan, H. Anion-exchange membrane water electrolyzers and fuel cells. Chemical Society Reviews, 2022, 51, 9620-9693. DOI: 10.1039/D2CS00038E.

(2) Záchenská, J.; Ábel, M.; Mičušík, M.; et al. Nickel-based nanocoatings on 3D Ni foam for zero-gap alkaline water electrolysis. Journal of Applied Electrochemistry, 2020, 50, 959-971. DOI: 10.1007/s10800-020-01448-7.

(2)

(max 10 pages)

☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization

(max 10 pages)